



SD-1.54 – Cam and Follower Apparatus

A vertical pillar contains the main spindle for each cam to be rotated, whilst a dial gauge has its anvil resting on the top edge of the cam.

PRODUCT CODE: SD-1.54

Categories: Educational, SD-1.54-Cam And Follower Apparatus, SV - Theory Of Machines, SV Series - Statics & Vibrations

Tags: Acceleration Diagram , Automatic Handling Machinery , Cam And Follower Apparatus , Cams , Constant Speed , Dwell , Followers , Internal Combustion Engine , Linear Displacement Against Rotation , Mechanisms , SD-1.54 , Simple Harmonic Motion , Velocity Diagram

DESCRIPTION

A vertical pillar contains the main spindle for each cam to be rotated, whilst a dial gauge has its anvil resting on the top edge of the cam.

Each cam is mounted on a central shaft and is free to rotate about this shaft. The cams are rotated by hand. The rise and fall displacement of the cam, as it rotates through 360°, is measured using the dial gauge.

The angular position of the cam is shown on a protractor and a pointer. Cams and followers are easily changed and graphs for the variety of cams can be created to compare the linear change with the rise and fall of the follower. A set of cams are supplied as standard along with a set of followers which run on the perimeter of each cam. Each cam can be easily changed and the angular and dial gauge readings are clear.

TECHNICAL SPECIFICATIONS

- **Simple harmonic motion cam:** Rise/Fall 24mm
- **Tangent cam:** Large radius 32mm; small radius 15mm
- **Eccentric cam:** 10mm eccentricity
- **Roller follower:** Ø19mm
- **Dial gauge:** 25mm range, 0.01mm resolution
- Angular protractor scale with 360° range and 1° resolution

FEATURES & BENEFITS

- Self-contained bench top unit



- Set of five cams supplied
- Set of two followers supplied
- Angular and linear displacement recorded
- Derive velocity and acceleration diagrams

Experimental Content

- To determine a graph of follower displacement against angular rotation of the cam
- To assess the effect of different cams and followers
- To derive the velocity and acceleration diagrams, and determine the maximum acceleration of the follower
- To compare experimental and calculated results

Ordering Specification

- 1 x SD-1.54 Assembly
- 1 x Simple harmonic motion cam
- 1 x Uniform motion cam
- 1 x Constant acceleration cam
- 1 x Tangent cam
- 1 x Eccentric cam
- 1 x Dial gauge
- Instruction manual
- Packing list
- Test sheet

Series Includes:

Theory of Machines

- SD-1.50 Wheel and Axle
- SD-1.51 Wheel and Differential Axle Apparatus
- SD-1.52 Efficiency of Screw Threads
- SD-1.53 Gear Tooth Form Apparatus
- SD-1.54 Cam and Follower Apparatus
- SD-1.55 Simple Flywheel
- SD-1.56 Epicyclic Gear Train
- SD-1.57 Castor, Camber, and King Pin Inclination
- SD-1.58 Relation Between Angular and Linear Speeds
- SD-1.59 Disc Brake Apparatus
- SD-1.60 Static & Dynamic Balancing Apparatus
- SD-1.61 Governor Apparatus
- SD-1.62 Gear Assembly Unit Combined Drives
- SD-1.63 Single Epicyclic Gear Train

Shipping Specifications



Volume: 0.14m³

Gross Weight: 18kg

Dimensions

Length: 0.25m

Width: 0.15m

Height: 0.54m

Order Codes

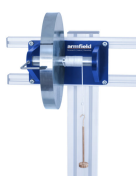
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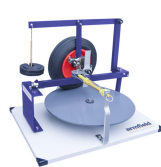
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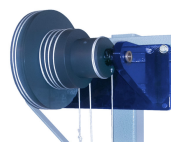
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